



Barista Skills Professional

Mandatory Activities Worksheet (中文)



咖啡师高级

Barista Professional

必修活动工作表

Mandatory Activities Worksheet

栏目SECTIONS	最长时间（分钟） MAXIMUM TIME (MINS)
1. 水化学评估 WATER CHEMISTRY EVALUATION	30
2. 填压和布粉练习 TAMPING & DISTRIBUTION EXERCISE	60
3. 简单的财务计算练习 SIMPLE FINANCIAL CALCULATION EXERCISE	30
总TOTAL	120分钟120 MINS

给学生的说明： GENERAL INSTRUCTIONS TO CANDIDATES:

- 这些练习是课程的必修部分，但不会通过考试进行测试。 These exercises form a mandatory part of your course but will not be tested by examination.
- 你应该在这些练习中评估自己的发现。没有规定的结果或预期结果。 You are expected to evaluate your own findings from these exercises. There are no prescribed outcomes or expected results.

1. 水化学评估 WATER CHEMISTRY EVALUATION

目的 Objective:

- 评估水化学（碱度和总硬度）对冲煮咖啡感官品质的影响 To evaluate the effects of water chemistry (alkalinity and total hardness) on the sensory qualities of brewed coffee.

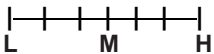
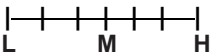
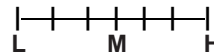
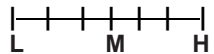
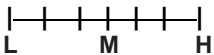
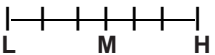
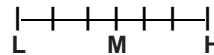
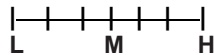
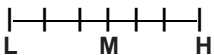
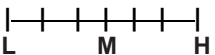
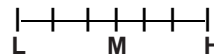
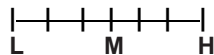
时间: 30分钟 Time: 30 minutes

设备: Equipment:

- SCA杯测协议中描述的设备可以在这里找到: <https://sca.coffee/research/protocols-best-practices> The equipment described in the SCA Cupping Protocol here:
- 另外, 还有3个手动水壶和一个精确的温度计 Plus, 3 manual kettles and an accurate thermometer

说明: Instructions:

- 从AST处分别获得3种不同类型的水每种750毫升（非常硬, 理想的SCA, 非常软/蒸馏水）。 Obtain from the AST 750mls each of 3 different water types (very hard, SCA ideal, very soft/distilled).
- 每种水保留约20毫升进行测试。 Retain approximately 20mls of each water for testing.
- 以固定的研磨大小并一次连续地为9个杯测碗研磨足够的咖啡。 Grind up adequate coffee for 9 cupping bowls at a fixed grind size and in a single, continuous batch.
- 每种水使用1 x水壶, 每种水冲泡3杯相同的咖啡, 并注意在恒定温度下输送每种水。 Using 1 x kettle per water type, brew 3 identical bowls of coffee, per water type, taking care to deliver each water at a consistent temperature.
- 品尝前, 从每套至少1碗中取出20毫升咖啡, 并留作测试。 Before tasting, withdraw 20mls of coffee from at least 1 bowl per set and retain for testing.
- 测试保留的水样品的临时和永久硬度, 并使用ppm CaCO₃当量标度在SCAE水表上作图（您的审查员将对此进行解释）。 Test the retained water samples for Temporary and Permanent Hardness and plot on the SCAE Water Chart, using a ppm CaCO₃ equivalent scale (your examiner will explain this).
- 测试保留的咖啡的TDS%和提取%。 Test the retained coffee for TDS % and Extraction %.
- 填写下面的感官和测量属性表。 Complete the Sensory & Measured Attributes table below.
- 填写下表后, 请完成“结论”部分。 Complete the Conclusion section after you fill in the table below.

感官和测得的属性 SENSORY & MEASURED ATTRIBUTES			
套装1 (水1) SET 1 (WATER 1)			
碱度 ALKALINITY	总硬度 TOTAL HARDNESS	TDS%	萃取 %EXTRACTION %
酸味 SOURNESS	甜味 SWEETNESS	苦味 BITTERNESS	醇厚度 BODY
			
其他描述 (例如渣感, 平, 厚重, 弱等) OTHER DESCRIPTORS (E.G.CHALKY, FLAT, HEAVY, WEAK ETC)			
套装2 (水2) SET 2 (WATER 2)			
碱度ALKALINITY	总硬度 TOTAL HARDNESS	TDS%	萃取% EXTRACTION %
酸味 SOURNESS	甜味 SWEETNESS	苦味 BITTERNESS	醇厚度 BODY
			
其他描述 (例如渣感, 平, 厚重, 弱等) OTHER DESCRIPTORS (E.G.CHALKY, FLAT, HEAVY, WEAK ETC)			
套装3 (水3)			
碱度 ALKALINITY	总硬度 TOTAL HARDNESS	TDS%	萃取 %EXTRACTION %
酸味 SOURNESS	甜味 SWEETNESS	苦味 BITTERNESS	醇厚度BODY
			
其他说明者 (例如胖, 平, 沉重, 弱等) OTHER DESCRIPTORS (E.G.CHALKY, FLAT, HEAVY, WEAK ETC)			

结论： Conclusion:

您注意到两组之间有任何区别吗？如果是这样，那是什么？ Did you notice any differences between the sets? If so, what were they?

碱度和永久硬度的影响是否与SCAE水表上的指示一致？ Was the effect of alkalinity and permanent hardness consistent with the indications on the SCAE Water Chart?

2. 填压和布粉评估 TAMPING & DISTRIBUTION EVALUATION

目的: Objective:

- 评估不同布粉方法对冲煮的意式浓缩咖啡的影响 Evaluate the effects of different distribution methods on brewed espresso.
- 评估不同的压粉方法对冲煮的意式浓缩咖啡的影响。 Evaluate the effects of different tamping methods on brewed espresso.

时间: 60分钟 Time: 60 minutes

设备: Equipment:

- 1个市售布粉工具 (例如OCD) 1 x commercially available distribution tool (e.g. OCD)
- 1 x正确安装的平面型压粉器 (通常为58.4mm直径) 1 x correctly fitting flat based tamper (usually 58.4mm diameter)
- 1 x不正确安装的平面型压粉器 (通常直径为57-58mm) 1 x incorrectly fitting flat based tamper (usually 57-58mm diameter)
- 1套范围超过40kgs / 90lbs的磅秤 (通常是浴室磅秤) 标准的意式浓缩咖啡生产设置 1 x set of scales with a range exceeding 40kgs/90lbs (usually bathroom scales) A standard espresso production setup
- 标准的意式浓缩咖啡测量设置 (折光仪和相关工具) A standard espresso measuring setup (refractometer and associated tools)
- 无底手柄portafilter (可选, 但对于视觉反馈很有用) Bottomless portafilter (optional, but useful for visual feedback)

说明: Instructions:

- 使用任何特定的压粉工具或布粉流程, 但流程必须是可重复的并且将成为您的参照冲煮。 Use any particular tamping tool or distribution routine, but routine must be repeatable and will be your control brew.
- 使用天平或任何其他准确的方法来找到一致的填压压力。 Use scales or any other accurate method to find a consistent tamp pressure.
- 对AST校准的配方/方法制作3杯双份浓缩咖啡。 Make 3 double espressos to the recipe/method calibrated by the AST.
- 测量每种冲泡的实际剂量/产量, 并计算出准确的EBF /比率。 Measure actual dose/yield for each brew and calculate accurate EBF/Ratios.
- 测试并品尝每种冲煮 Test and taste each brew.
- 计算每个参数的平均平均值和标准偏差 (SD) (您的培训师会帮助您解决, 并解释为什么这些是有用的数据点) 。 Calculate mean averages and standard deviation (SD) for each parameter (your trainer will help with this and explain why these are useful data points).
- 将结果记录在下面“结果”部分的表1: 对照配方/方法中。 Record the results in Table 1: Control Recipe/Method in the Results section below.

除非培训师另有指示，否则

您只能从以下选择中选择一个实验：A或B

Unless Otherwise Instructed by Your Trainer,
You May Choose Only One Experiment: A Or B, From The Choices Below

A. 布粉方法测试 Distribution Method Test

- 分组确定要评估的两种不同的布粉方法。（例如，水平敲击与手指分布）。As a group, decide which two different distribution methods you would like to evaluate.(E.G horizontal settling vs finger strike distribution).
- 应使用磅秤来确保始终一致的填压压力。You should use the scales to ensure consistent tamp pressure throughout.
- 使用适合的填压器和秤，使用第一种布粉方法作为对照配方准备3组双份浓缩咖啡。Using the fitting tamper and the scales, prepare 3 double espressos to the control recipe, using your first distribution method
- 将结果记录在表2中：测试方法1Record the results in Table 2: Test Method 1 below.
- 通过使用合适的填压器和秤，可以使用第二种布粉方法作为对照配方准备3组双份浓缩咖啡。Using the fitting tamper and the scales, prepare 3 double espressos to the control recipe, using your second distribution method.
- 将结果记录在下面的表3：测试方法2。Record the results in Table 3: Test Method 2 below.
- 填写“结果”表后，请完成“结论”部分。Complete the Conclusion section after you fill in the Results table.

B. 填压方法测试 Tamping Method Test

- 作为一个小组，决定要评估哪种两种填压方法。（例如，适合与不适合，轻与重，研磨，敲击和填压与研磨和填压，平底比对非平底）。As a group, decide which two different tamping methods you would like to evaluate.(E.G fitting vs non-fitting, light vs heavy, grind, settle and tamp vs grind and tamp, flat base vs non-flat base).
- 除非要测试填压压力变化，否则应使用磅秤确保整个填压压力一致。You should use the scales to ensure consistent tamp pressure throughout, unless you are testing tamp pressure variation.
- 使用第一种填压/压粉方法，准备3杯浓缩意式浓缩咖啡作为对照配方。Prepare 3 double espressos to the control recipe, using your first tamper/tamping method.
- 将结果记录在下面的表2：测试方法1中。Record the results in Table 2: Test Method 1 below.
- 使用第二种粉锤/填压方法，准备3杯浓缩意式浓缩咖啡作为对照配方。Prepare 3 double espressos to the control recipe, using your second tamper/tamping method.
- 将结果记录在下面的表3：测试方法2中。Record the results in Table 3: Test Method 2 below.
- 填写“结果”表后，请完成“结论”部分。Complete the Conclusion section after you fill in the Results table.

结果: Results:

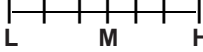
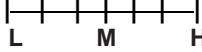
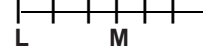
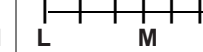
	表1: 参照配方/方法 TABLE 1: CONTROL RECIPE/METHOD			
	EBF%	RATIO 比	TDS %	EXT %
冲煮1 BREW 1				
冲煮2 BREW 2				
冲煮3 BREW 3				
平均AVERAGE				
标准差 SD				
	酸度 ACIDITY	甜味 SWEETNESS	苦味 BITTERNESS	醇厚程度 BODY
冲煮2 BREW 2 (只要) (ONLY)				

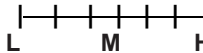
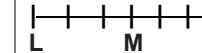
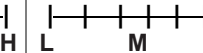
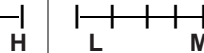
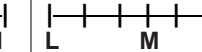
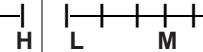
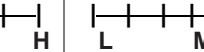
	表2: 测试方法1 TABLE 2: TEST METHOD 1			
	EBF%	RATIO 比	TDS %	EXT %
冲煮1 BREW 1				
冲煮2 BREW 2				
冲煮3 BREW 3				
平均AVERAGE				
标准差 SD				
	酸度 ACIDITY	甜味 SWEETNESS	苦味 BITTERNESS	醇厚程度 BODY
冲煮2 BREW 2 (只要) (ONLY)				

	表3: 测试方法2 TABLE 3: TEST METHOD 2			
	EBF%	RATIO 比	TDS %	EXT %
冲煮1 BREW 1				
冲煮2 BREW 2				
冲煮3 BREW 3				
平均AVERAGE				
标准差 SD				
	酸度 ACIDITY	甜味 SWEETNESS	苦味 BITTERNESS	醇厚程度 BODY
冲煮2 BREW 2 (只要) (ONLY)				

结论： Conclusion:

在将测试方法1的结果与对照方法的结果进行比较时，您是否注意到任何明显的趋势？

如果是，您注意到了什么？ Did you notice any clear trends when comparing the results of test method 1 to your control method? If yes, what did you notice?

在将测试方法2的结果与对照方法进行比较时，您是否注意到任何明显的趋势？ 如果

是，您注意到了什么？ Did you notice any clear trends when comparing the results of test method 2 to your control method? If yes, what did you notice?

在比较测试方法1和测试方法2的结果时，您是否注意到任何明显的趋势？ 如果是，您注意到

了什么？ Did you notice any clear trends when comparing the results of test method 1 to test method 2? If yes, what did you notice?

3. 简单的财务计算 SIMPLE FINANCIAL CALCULATIONS

目的: Objective

- 了解如何在确定菜单项价格时使用成本和销量。 Understand how costs and sales volumes are used in determining the price of menu items.

时间: 30 分钟 Time: 30 Minutes

A 部分: 确定已售商品成本 Determining Cost of Goods Sold

场景: Scenario:

- 有一家名为“Only Lattes”的咖啡馆。 There is a café called “Only Lattes.”
- 这家咖啡馆仅出售一种菜单产品，即 12 盎司/ 355 ML 的热咖啡拿铁。 This café only sells one menu item which is a 12-ounce/355 ML hot café latte.
- 这杯拿铁的售价为 5 欧元（不含税）*。 This latte is sold for €5 plus tax*.

说明: Instructions:

- 在下表中，前两列包含“Only Lattes”的费用，按单位细分。
In the table below, the first two columns contain some of Only Lattes' costs, broken down by unit.
- 分组讨论并确定哪些可用成本应包含在“已售商品成本”列中
As a group discuss and identify which available costs should be included in the cost of goods sold column
- 将这些成本合计以确定所售商品的成本 Total these costs to determine the cost of goods sold

已知支出 Available Costs	€	销货支出 Cost of Goods Sold (可变支出) (variable costs)	€
咖啡熟豆 Roasted coffee	0.4		
电力 Electricity	1.2		
纸杯 Paper cup	0.06		
会员 Lid	0.02		
租金 Rent	2		
牛奶 Milk	0.4		
直接人工 Direct Labor	.3		
保险 Insurance	.05		
会计费用 Accounting Fees	.05		
		销货成本总计 Cost of Goods Sold Total:	

*税款是代表政府收取的，不属于税收（销售），因此请在不计税的情况下进行计算（例如增值税等）。*Tax is collected on behalf of the GOVERNMENT and is NOT part of the Revenue (Sales) so calculate all without Tax (e.g. VAT etc.).

出售商品的成本占售价的百分之几? The cost of goods sold is what percentage of the selling price?

B 部分：基于设定价格确定收支平衡的销售量 Determining Breakeven Sales Volume Based on a Set Price

场景： Scenario:

- Only Lattes 咖啡厅已经成功发展，并搬入了一座较大的建筑物，其租金和水电费更高。 The café Only Lattes has grown successfully and has moved into a larger building with higher cost in rent and utilities.
- 咖啡厅老板不想提高热咖啡拿铁的价格。 The café owners do not want to raise the price of the hot café lattes.

说明： Instructions:

- 下面是咖啡馆新的固定费用表。 Below is a table of the café's new fixed costs.
- 使用上面计算出的售出商品成本，作为一个小组，确定每杯拿铁的销售收入可以转化为下表中列出的固定成本。 Using the cost of goods sold calculated above, as a group determine how much revenue from the sale of each latte can go toward the fixed costs listed in the chart below.
- 根据得出的计算结果，小组确定必须卖出多少拿铁才能覆盖图表中列出的固定费用。 Based on the resulting calculation, as a group determine how many lattes would have to be sold to also cover the fixed expenses listed in the chart.

每月固定成本 Monthly Fixed Costs	€
租金 Rent	3,000
设备租赁 Equipment leases	1,500
会计费用，保险，公用事业等 Accounting fees, Insurance, Utilities etc.	500
总： Total:	5,000(€)

$$\begin{array}{l}
 \text{单价€5-} \underline{\hspace{2cm}} = \underline{\hspace{2cm}} \\
 \text{Unit Price €5 - } \underline{\hspace{2cm}} = \underline{\hspace{2cm}} \\
 \hspace{10em} (\text{COGS}) \hspace{10em} (\text{Contribution Margin}) \\
 \hspace{10em} (\text{COGS}) \hspace{10em} (\text{边际收益})
 \end{array}$$

5,000 欧元的固定成本

$$\begin{array}{l}
 \text{Fixed Costs of €5,000 / } \underline{\hspace{2cm}} = \underline{\hspace{2cm}} \\
 \hspace{10em} (\text{边际收益 Contribution Margin}) \hspace{10em} (\text{收支平衡量 Break Even Sales Volume})
 \end{array}$$